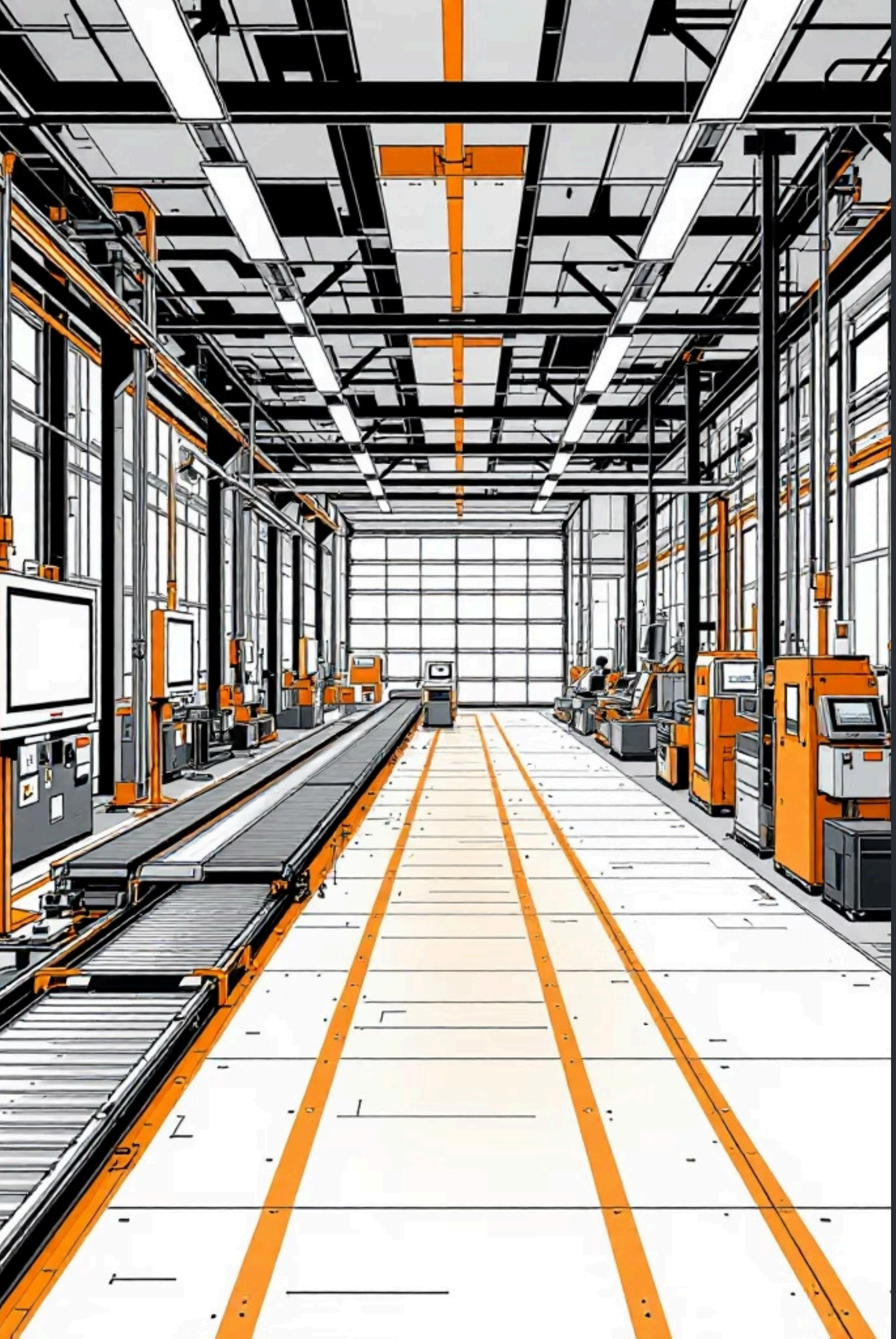


Sahyog Industries: Digital Transformation Roadmap

2025–2026 | Comprehensive Programme Plan

CONFIDENTIAL | ACCREDIAN CAPSTONE PROJECT

Executive Summary

Sahyog Industries — a 45-year-old mid-sized Indian manufacturer — faces urgent pressure from rising competition, supply-chain disruptions, and demand for lean, customised production. This roadmap covers strategy, people, process, and technology to achieve digital maturity within 12 months.

Rs. 4.2Cr	~172%	12M	Lvl 4
Year 1 Budget	Projected ROI (Yr 2)	Programme Duration	Target Maturity
		4 Phases	From Level 2

Programme Ambition: Transform Sahyog from a digitally-emerging manufacturer (Level 2) to an integrated, data-driven enterprise (Level 4), delivering **20–25% operational efficiency gains** and **15–20% revenue growth** with positive ROI by Q4 Year 1.

Strategic Alignment & Digital Maturity

Five Strategic Objectives

Operational Efficiency

IoT, predictive maintenance — 20–30% downtime reduction

Supply Chain Resilience

AI forecasting, real-time tracking — lead times cut from weeks to days

Customisation & Flexibility

ERP + digital twin — 3x more custom SKUs; 15% higher order value

Market Competitiveness

B2B portal + analytics — 15–20% revenue growth

Innovation & Sustainability

Energy monitoring, waste analytics — 10% energy cost reduction

Digital Maturity Gaps (Current → Target)

Dimension	Now	12M Target
Manufacturing & Operations	Level 1–2	Level 3–4
Supply Chain & Logistics	Level 2	Level 4
Customer & Sales	Level 1–2	Level 3
Data, Analytics & Intelligence	Level 1	Level 3
Governance, Security & Compliance	Level 2	Level 4

Overall current maturity: **Level 2 (Emerging)**. Critical gaps in data infrastructure, customer-facing tools, and advanced technology adoption.

SWOT: Digital Transformation Readiness

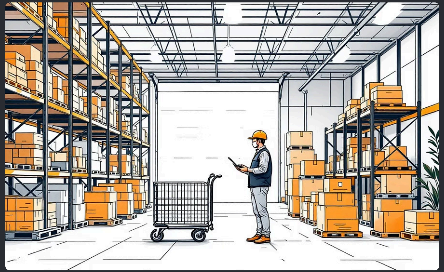
- **Strengths -**
  - Sahyog's 45-year legacy
  - established customer base, distribution network, skilled workforce
- **Weaknesses -**
  - legacy systems, manual processes,
  - limited IT infrastructure, and absence of modern CRM
- **Opportunities -**
  - lean manufacturing demand, Industry 4.0 push
  - IoT/blockchain advantages for new revenue
- **Threats -**
  - digital competitors, supply chain disruptions,
  - customer expectations, and cybersecurity/privacy risks

Customer & Market Strategy

Four key segments drive Sahyog's digital value proposition:

B2B OEM Buyers

Self-service portal: custom configuration, order tracking, digital invoicing, SLA dashboard



National Distributors

Real-time stock levels, automated replenishment, performance analytics

Export / International Buyers

Blockchain traceability, digital quality certificates, export compliance automation

Plant Managers

IoT dashboards, predictive maintenance alerts, OEE monitoring, shift performance reports

Technology Architecture & Digital Stack

Priority Use Cases

1

Predictive Maintenance

IoT + Edge AI — reduce unplanned downtime 30–40% **(Q1–Q2)**

2

B2B Self-Service Portal

Cloud + CRM/ERP API — cut manual order handling 60% **(Q1–Q2)**

3

ERP Implementation

SAP S/4HANA or Oracle NetSuite — end-to-end integration **(Q1–Q3)**

4

Demand Forecasting

AI/ML + Data Lake — reduce inventory holding costs 20–25% **(Q2–Q3)**

Implementation Strategy: Technology Architecture

Deploying Sahyog's five-layer digital stack requires a strategic, phased approach, aligning technology implementation with business value and talent development over a 12-month roadmap.

1

Q1: Foundational Layers

Layers: Edge & IoT, Core Enterprise (Planning)

- **Milestones:** IoT sensor deployment (pilot), SCADA/PLC integration, ERP/MES vendor selection & blueprint. Cloud partner selection.
- **Teams:** Operations, IT Infrastructure, Procurement.
- **Success:** Real-time operational data streams established, core system requirements defined.

2

Q2: Integration & Data Backbone

Layers: Integration & Data, Core Enterprise (Initial Deployment)

- **Milestones:** API Gateway setup, Data Lake implementation, ERP/MES module go-live (e.g., Inventory, Production).
- **Teams:** Data Engineering, IT Development, Business Process Owners.
- **Success:** Data centralisation, automated data flow, first phase of core systems operational.

3

Q3: Advanced Analytics & Process Optimization

Layers: Analytics & AI/ML, Core Enterprise (Full Deployment)

- **Milestones:** Power BI dashboards for operations, predictive maintenance model (pilot), ERP/MES full system integration (WMS), CPQ/CRM selection.
- **Teams:** Data Science, Business Intelligence, Business Units.
- **Success:** Data-driven decision making, reduced downtime, enhanced operational visibility.

4

Q4: Customer-Centric & AI-Driven

Layers: Customer & Sales, Advanced Analytics & AI/ML

- **Milestones:** B2B self-service portal launch, CRM/distributor portal deployment, AI-driven demand forecasting, cyber security
- **Teams:** Sales, Marketing, Customer Service, Cybersecurity.
- **Success:** Improved customer experience, optimized inventory, robust security posture.

Change Management: Driving Adoption

Successful digital transformation at Sahyog Industries is not just about technology; it's fundamentally about people. Proactive change management is crucial to ensuring seamless adoption, mitigating resistance, and fostering a digitally-enabled, future-ready workforce.

These five pillars work as a reinforcing change system: executive direction sets the tone, communication builds trust, training builds capability, engagement creates ownership, and culture sustains adoption.

The change impact spreads from leadership to managers to employees, with feedback loops that help reduce resistance and accelerate adoption.

Change Management: Five Pillars of Success

Leadership & Vision

Secure unwavering executive sponsorship and articulate a clear, compelling digital vision that inspires and guides the entire organisation towards new ways of working.

Strategic Communication

Implement a multi-channel communication plan for transparent, frequent updates, addressing concerns, and celebrating milestones to build trust and momentum.

Training & Upskilling

Develop comprehensive training programmes tailored to different roles and digital tools, ensuring employees gain necessary skills and confidence for the transformed environment.

Stakeholder Engagement

Actively involve employees from all departments, suppliers, and key customers early on to co-create solutions and foster a collective sense of ownership.

Culture & Adoption

Cultivate a culture that embraces innovation, continuous learning, and data-driven decision-making, recognising and rewarding early adopters and digital champions.

Critical success factors: visible executive sponsorship, frequent communication, role-based training, early stakeholder involvement, and recognition of digital champions.

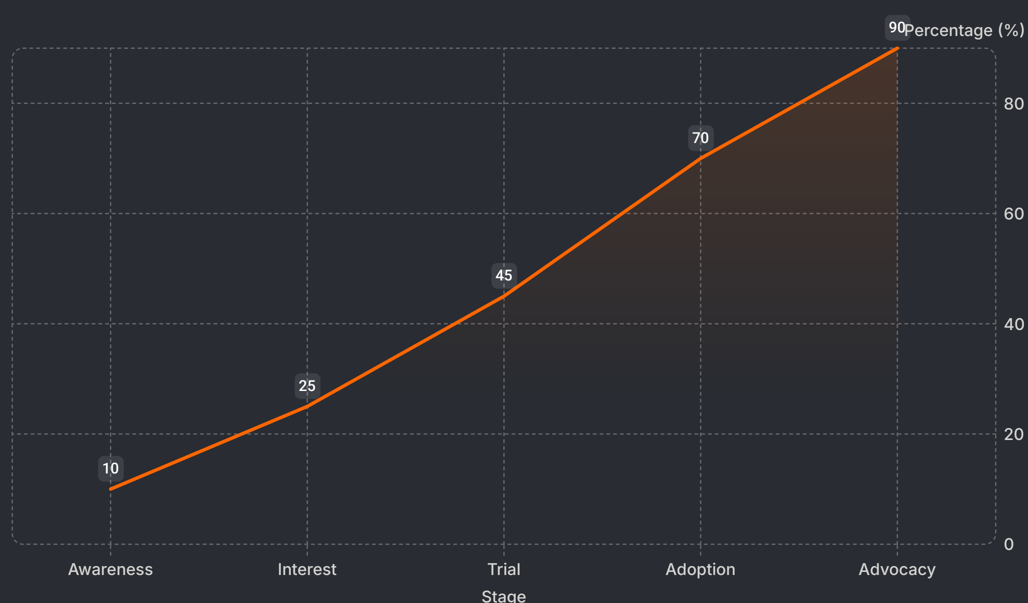
Milestones: leadership alignment, pilot group readiness, first training completion, early adoption sign-off, and enterprise-wide reinforcement.

Adoption Metrics & Resistance Management

Successful digital transformation hinges on understanding and actively managing both the adoption curve of new technologies and the inevitable resistance encountered during change. Measuring these metrics allows us to proactively address challenges and accelerate the transition to new ways of working.

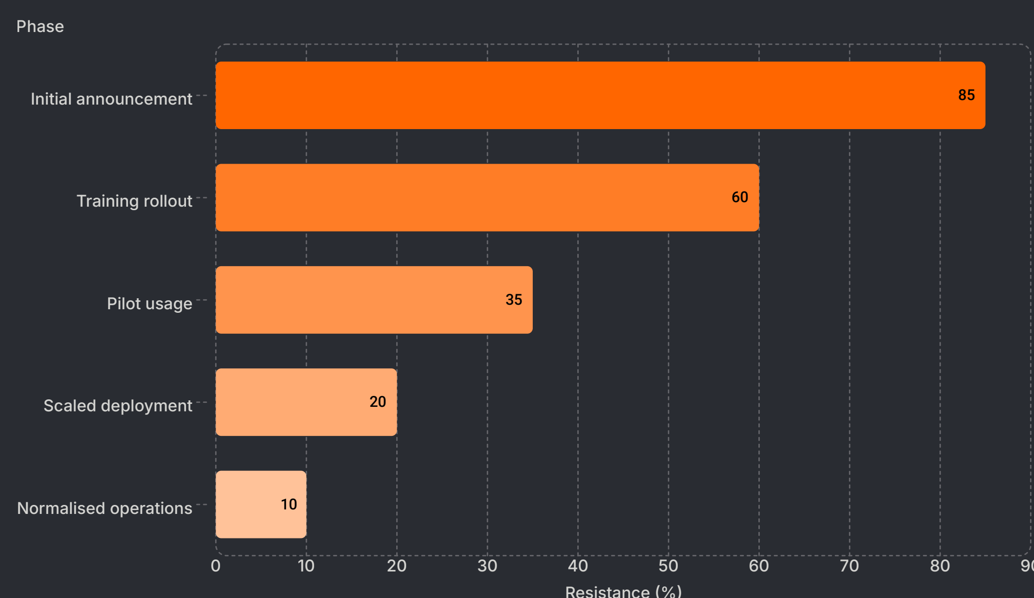
Adoption Curve

Adoption starts with a small group of champions, expands through early adopters, and reaches the wider organisation once confidence and familiarity increase.



Resistance Management

Resistance is highest at the start and declines as communication, training, and support increase.



12-Month Phased Roadmap

1

Q1: Foundations

Digital Office setup; ERP vendor signed; IoT PoC on 1 line; cloud infra & data lake built

2

Q2: Pilot & Validate

ERP pilot go-live; B2B Portal v1 launched; IoT live on 2 lines; core team training complete

3

Q3: Scale & Expand

ERP rolled out to all plants; predictive maintenance live; demand forecast v1 deployed

4

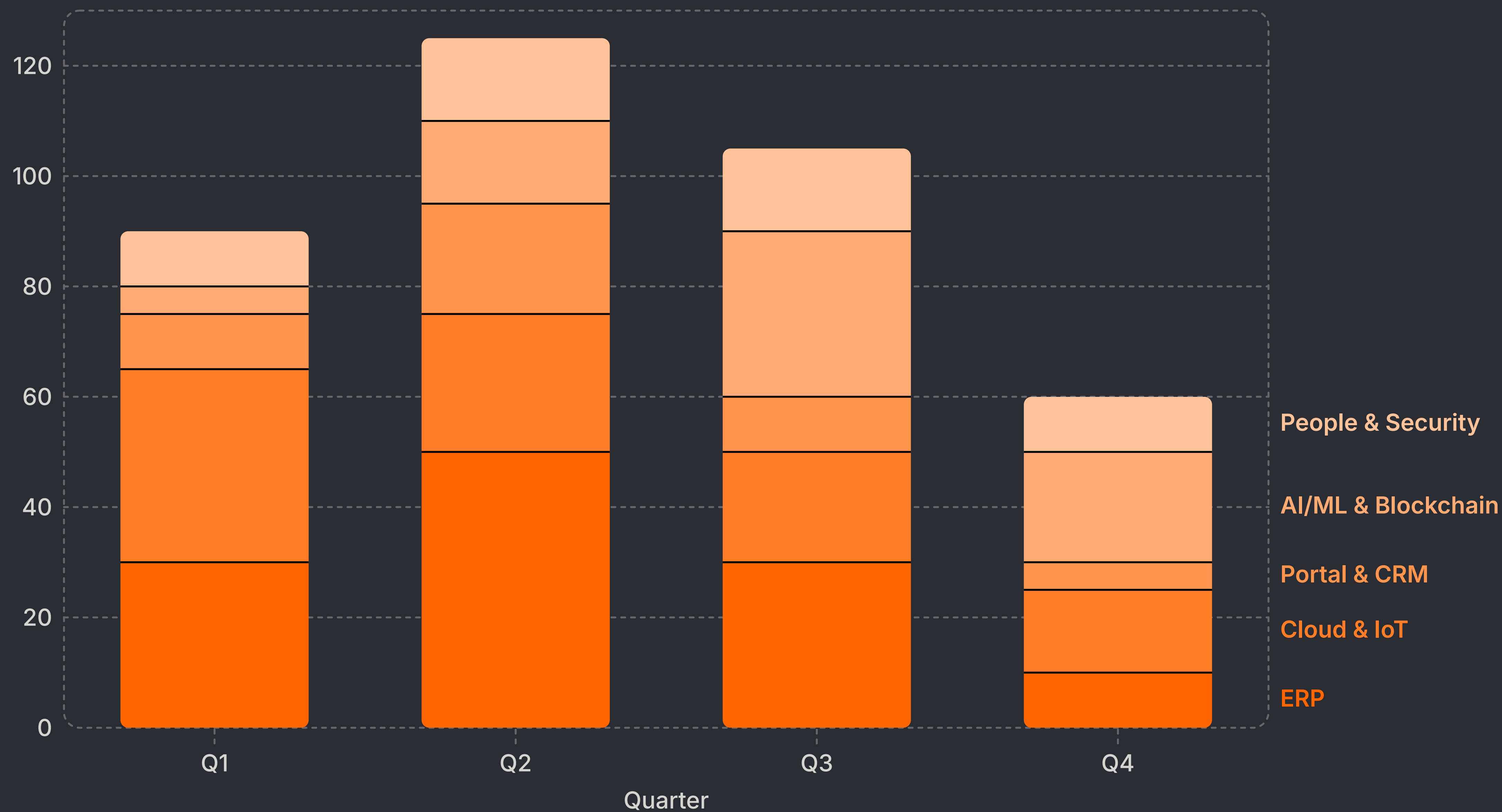
Q4: Optimise & ROI

Blockchain live; digital twin PoC; full analytics suite; ROI report delivered to Board

Four quarters build in sequence from foundations to pilot, then scale, and finally optimization and ROI.

Budget Estimates & Investment Breakdown

The digital transformation initiative requires a significant investment over the next year, strategically allocated across key technology pillars to drive maximum business value. This phased budget ensures resources are aligned with our implementation roadmap.



The stacked column chart illustrates the Year 1 budget allocation by quarter, measured in Rs. Lakhs. Investments are front-loaded for ERP and Cloud/IoT infrastructure in Q1 and Q2, while AI/ML and Blockchain initiatives see increased funding in Q3 and Q4 as foundational layers become operational. People and Security remain a consistent investment throughout the year.

Investment Summary

Total Year 1: Rs. 4.20 Crore

This includes Rs. 3.80 Cr for implementation efforts and a Rs. 0.40 Cr contingency fund to manage unforeseen challenges.

ERP: Rs. 1.20 Cr

Represents the largest single investment, strategically phased across all four quarters to support a comprehensive SAP S/4HANA or Oracle NetSuite implementation.

IoT & Cloud: Rs. 0.95 Cr

Dedicated to establishing the edge infrastructure, deploying IoT sensors, and building the Azure Data Lake for robust data collection and processing capabilities.

People & Security: Rs. 0.60 Cr

Allocated for crucial talent development, comprehensive change management programs, and ensuring compliance with data protection and privacy (DPDP) regulations.

Risk, Security & Compliance

Integrated risk management, cybersecurity controls, and compliance governance to protect delivery, operations, and customer trust.

Top Risks & Mitigations

ERP Overrun

Severity: High | **Probability:** Medium

Fixed-price contracts; phased rollout; 10% contingency budget

Budget Overrun

Severity: Medium | **Probability:** High

Change control board; MoSCoW prioritisation; monthly Steering Committee reviews

Change Resistance

Severity: High | **Probability:** High

Change champions programme; executive sponsorship; incentive alignment from Day 1

Cybersecurity Breach

Severity: Critical | **Probability:** Medium

ISO 27001 roadmap; quarterly VAPT; Zero Trust architecture; incident response drills

DPDP Non-Compliance

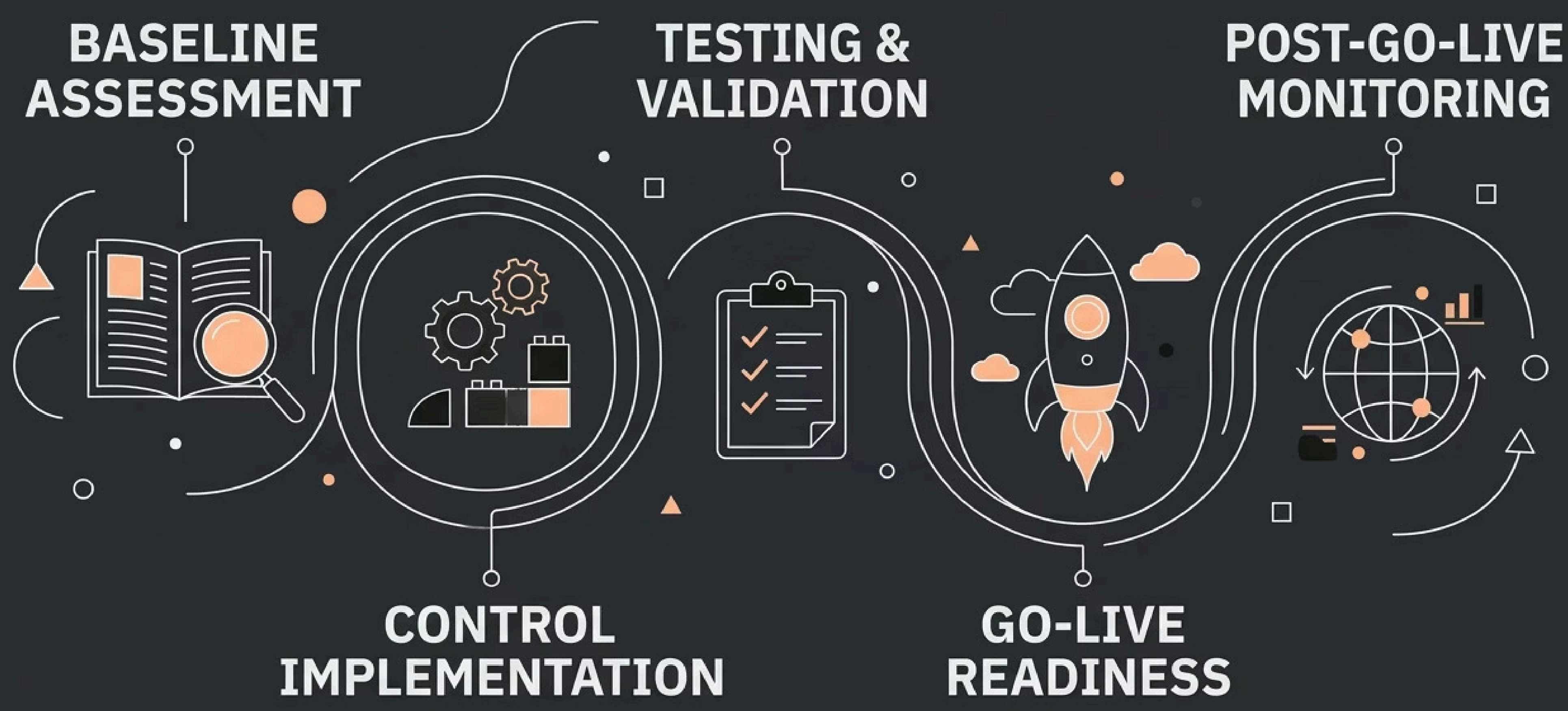
Severity: High | **Probability:** Medium

Appoint DPO; consent management system; legal counsel engagement

Compliance Roadmap & Controls

Integrated risk management, cybersecurity controls, and compliance governance to protect delivery, operations, and customer trust.

Compliance Roadmap Milestones



This visual roadmap outlines the sequential phases for achieving and maintaining robust compliance, ensuring that each step builds upon the previous one to secure our digital assets and operations.

Critical Security Controls

- Zero Trust access, MFA are mandatory for privileged users
- Quarterly VAPT and incident response drills are required for ongoing assurance

Compliance Requirements

- DPO appointed, consent records maintained, and breach notification within 72 hours
- India-region data localization and legal review embedded into the operating model

Cybersecurity & Compliance Framework

Network Security

- Next-gen firewalls
- VLAN OT/IT segmentation
- IDS/IPS
- DDoS protection

Identity & Access

- RBAC
- MFA enforced
- Privileged Access Management (PAM)

Data Security

- AES-256 at rest
- TLS 1.3 in transit
- DLP tools
- Geo-redundant backups

DPDP Act 2023

- Consent management
- DPO appointed
- 72-hr breach notification
- India-region data localisation

Key Performance Indicators (KPIs)

Performance targets define how the transformation will improve operational efficiency, customer experience, and digital growth. These KPIs track progress against the 12-month target state and show the expected impact across the business.

Key Performance Targets

Category	KPI	Baseline	Target (12M)	Impact
Operational	OEE	52%	**68%**	↑ +16 pts
Operational	Unplanned Downtime	48 hrs/mo	**<20 hrs/mo**	↓ -58%
Operational	OTIF Delivery	68%	**88%**	↑ +20 pts
Operational	Inventory Holding Days	45 days	**28 days**	↓ -17 days
Customer	NPS	32	**52**	↑ +20
Customer	B2B Portal Adoption	0%	**70%**	New channel
Customer	Order Visibility	Manual	**Real-time**	Faster service
Financial	Revenue from Digital	0%	**20% of total**	Growth lever

Performance outlook: Operational gains drive efficiency, customer metrics improve service quality, and digital revenue creates a new growth engine.

4.20Cr

Total Investment

Total programme investment

3.40Cr

Gross Benefit

50% realised in Year 1

172%

Year 2 ROI

On cumulative realisation

Operational uplift

Higher OEE, lower downtime, and stronger delivery reliability

Customer experience

Better visibility, adoption, and service responsiveness

Digital growth

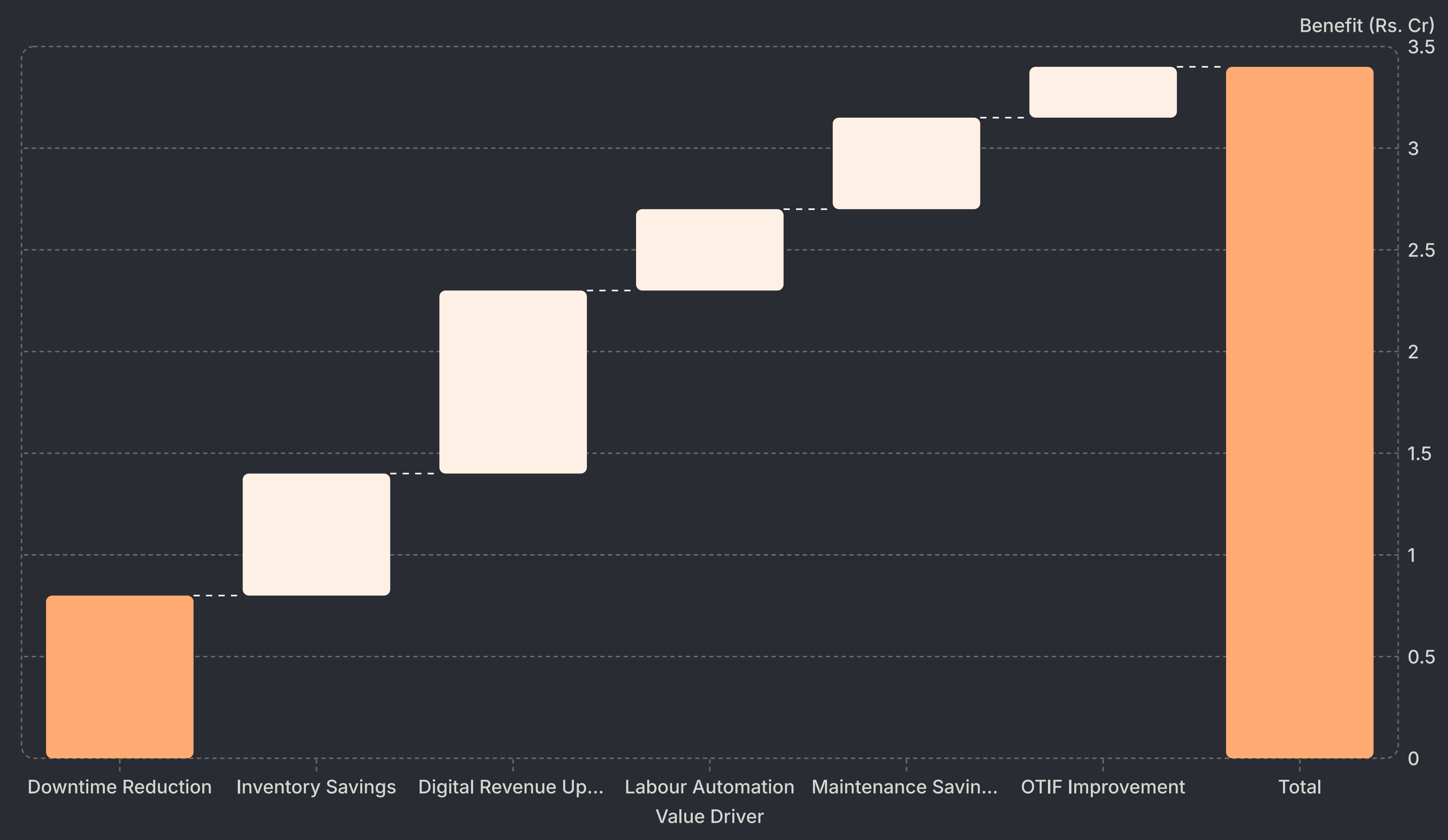
New revenue streams and measurable transformation returns

ROI & NPV Analysis

This section details the expected financial returns and value creation from the digital transformation initiative, projecting the significant impact on profitability and long-term enterprise value.

ROI Summary

The digital transformation is projected to generate substantial annual benefits from multiple value drivers, leading to a strong return on investment.



Total Gross Benefit: Rs. 3.40 Cr (50% realisation in Year 1)

Year 2 ROI: ~172%

Break-even: Q2 Year 2

This initiative is set to deliver significant financial gains, with Year 2 ROI exceeding initial investments and a rapid break-even point.

NPV Analysis

1

Year 1

Partial benefit realisation begins; initial investment deployed.

2

Year 2

Payback achieved; positive cash flow accelerates.

3

Year 3

Full benefits realised; sustained value creation.

Year 1	Partial benefit	0.91	PV Positive
Year 2	Payback reached	0.83	PV Positive
Year 3	Full benefit	0.75	PV Positive

The 3-year Net Present Value (NPV) at a 10% discount rate is projected to be **+Rs. 7.2 Crore**, demonstrating strong long-term value creation.

The positive NPV confirms the project's financial viability and its potential to deliver substantial value to the organization over the next three years.

Benefit Realisation & Scenario View

01

Ramp-up

Initial implementation and partial benefit capture.

02

Payback

Cumulative benefits offset initial investment.

03

Stabilisation

Full benefit realisation and sustained operational improvements.

Scenario planning indicates that even with a 6-month delay in benefit realisation, the project maintains a positive NPV, albeit at a reduced value of +Rs. 3.5 Cr.

The Path Forward

Sahyog Industries is positioned to become a **digitally integrated, customer-centric manufacturer** — leveraging 45 years of domain expertise with the power of Industry 4.0.

Start Now

Digital Office live in Q1; IoT PoC and ERP vendor signed within 90 days

Bring Everyone Along

Change champions, role-based training, and executive sponsorship from Day 1

Measure Relentlessly

Monthly KPI dashboards; 70% portal adoption; ROI report delivered Q4

Stay Compliant

DPDP Act readiness, ISO 27001 roadmap, and Zero Trust security from the ground up

3-Year NPV at 10% discount rate: +Rs. 7.2 Crore | Break-even by Q2 Year 2 | Prepared by the Digital Transformation Lead Team | Accredian Capstone Project 2025